SCIENTIFIC AUDIT & PIPELINE TRANSFORMATION REPORT

Compound Sustainability Stress Index (SSI)

Comprehensive Technical Report of Changes from v1 to Canonical v2 Architecture

Date: October 14, 2026 **Repository:** Compound-Sustainability-Stress-Modeling **Git Base:** `ff880cdf85`

Reference Geographies: Mumbai (707 hexes) & Coimbatore (995 hexes)

Pipeline Verification: `make reproduce` (Exit Code 0, 34/34 tests)

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1. Executive Summary: Why the Overhaul Was Essential

The **Compound Sustainability Stress Index (SSI)** is a multi-dimensional geospatial framework designed to quantify urban environmental hazards (extreme heat, hydrological drought, atmospheric $PM_{2.5}$, vegetation depletion, and socio-economic exposure) at the neighborhood scale using the Uber H3 discrete global grid (resolution 8, ~0.74 km² per cell).

An exhaustive scientific audit of the original version (v1) revealed critical structural distortions that compromised the physical validity and policy utility of the modeling outputs:

> **Primary Failure Modes Identified in v1:**
>
> - **Seasonality Erasure:** Normalizing indicators independently within each month forced every city-wide monthly mean to exactly 0.50, subtracting out peak summer heatwaves, winter air pollution spikes, and monsoon rainfall relief.
> - **PCA Weighting Collapse:** PCA was fitted across heterogeneous indicators where PC1 captured only 34.56% of variance, heavily weighting atmospheric swings while reducing spatial exposure to negligible levels.
> - **Circular Testing:** Vulnerability was purported to be confirmed against building density ($R^2 \approx 0.97$), but building density was already an input to the vulnerability formula.
> - **Unstable Archetypes:** K-Means clustering produced 6 monthly regimes rather than stationary spatial typologies; individual hexes cycled through all 6 labels across calendar months.
> - **Air Quality Unit Inconsistencies:** $PM_{2.5}$ inputs exhibited unit ambiguities and lacked documented provenance.

Over a 14-step implementation program (Steps 02 through 15), the codebase was refactored from the ground up into **SSI v2**. The v2 release establishes a unified frozen global baseline, adopts a pre-registered primary equal-weighting scheme, implements non-linear smooth compound coupling, installs a pre-registered can-fail validation suite, discovers a stationary hex-level typology (k=2, ARI = 1.000), and packages the entire workflow into a byte-for-byte reproducible pipeline.

2. Side-by-Side Methodological Matrix (v1 vs. v2)

Dimension	Original v1 Framework	Canonical v2 Architecture
Baseline Normalization	Per-month local z-score clamping. Normalized against month-specific min/max or mean/std.	**Single global frozen reference baseline** (2nd–98th percentiles across pooled multi-city reference panel).
Physical Seasonality	**Erased by construction.** Every month centered at 0.50 (between-month variance share ≈ 0.00).	**Fully preserved.** Real annual cycles: May heat surge, July monsoon washouts, January pollution peaks.
Weighting Methodology	Unstable PCA PC1 loadings adjusted with ad-hoc city multipliers (R^2 = 1.000000 exact linear fit).	**Equal weighting (primary, w=0.2000)**. Compared against EWM, CRITIC, and PCA. High rank stability ($\rho \geq 0.9425$).
Compound Hazard Coupling	Multiplicative or hard thresholding (γ=0.30, 75th percentile) without sensitivity sweeps.	**Smooth interactive coupling:** ssi_v2 = ssi_lin + $\gamma\ \Omega\ (1 - \text{ssi_lin})$ (q=0.90, γ=0.30). Tested on 48-run grid.
$PM_{2.5}$ Pollution Data	Coarse annual SEDAC layers with unit and scaling ambiguities.	**Monthly CAMS EAC4 surface reanalysis** with strict 10^9 conversion ($kg/m^3 \rightarrow \mu g/m^3$). Verified via ADS.
Model Validation	**Circular:** Regressing vulnerability on its own building density input ($R^2 \approx 0.97$).	**Pre-registered can-fail protocol (V1–V6):** 2020 lockdown drop, 2022 heatwave, MODIS LST, Moran's I, held-out layers.
Urban Typology	6 transient monthly clusters; hexes cycled across labels every month (no actionable spatial zones).	**2 stationary hex archetypes** (k=2, silhouette = 0.5054, ARI = 1.000). 100% spatial contiguity coherence.
Cross-City Portability	Ad-hoc per-city models; parameters differed across cities, making comparisons invalid.	**Unified frozen parameter file:** Identical baseline cuts applied across Mumbai and Coimbatore.
Reproducibility	Manual scripts, overwriting files, undocumented intermediate data stores.	**End-to-end automated:** `make reproduce` runs pipeline from fit to figures in < 45s with exit code 0.

3. Detailed Implementation Breakdown (Steps 02 to 07)

Step 02 Provenance Audit & v1 Freeze

Before modifying production code, the complete state of v1 was audited and frozen. All v1 panels, intermediate files, and reports were copied to `archive/v1/`, and a cryptographic manifest (`archive/v1/MANIFEST.sha256`) was generated. Regression analysis in `audit/fit_check.py` reproduced the exact linear weights of v1 (R^2 = 1.000000): Heat (0.1714), Water (0.1452), Vegetation (0.1260), Pollution (0.2784), and Vulnerability (0.2791). A definitive repository Name Map was codified in `audit/provenance.md`.

Step 03 Scaffolding & Structured Results I/O

Standardized data contracts were established in `config.yml` under a dedicated `v2:` block, locking random seed `20260101`, designating Mumbai and Coimbatore as reference cities, and pre-registering primary parameters. The utility `src/utils/results_io.py` was implemented to enforce mandatory provenance metadata (Git commit SHA, SHA256 config hash, ISO timestamp) on every JSON and Parquet output. The validator `audit/schema_check.py` was created to enforce schema compliance.

Step 04 Raw Physical Driver Panels (`v2_raw`)

To decouple feature engineering from normalization, raw environmental observations were ingested and preserved strictly in physical units in `data/h3_panel/v2_raw/`:

- `heat_raw`: Heat Index (°C) combining ERA5 2m dry-bulb temperature and dewpoint temperature.
- `water_raw`: Precipitation deficit below normal + volumetric soil water layers 1 and 2 (mm / m^3/m^3).
- `veg_raw`: Sentinel-2 NDVI canopy deficit (unitless NDVI, inverted so higher = more degraded).
- `exposure_raw`: Composite of static Census 2011 poverty/slum/elderly shares and OpenStreetMap density.

The complete dictionary of raw and held-out evaluation variables was documented in `docs/data_dictionary.md`.

Step 05 CAMS Atmospheric $PM_{2.5}$ Ingestion

The CAMS EAC4 global reanalysis dataset was integrated in `src/ingest/cams_pm25_ingest.py`. Missing historical periods (December 2021) were retrieved directly from the Copernicus Atmosphere Data Store (ADS). To resolve past unit ambiguities, a unit conversion of 1.0×10^9 was applied, transforming raw mass mixing ratios (kg/m^3) into standard concentrations ($\mu g/m^3$). Verified against known physical standards (2.5×10^{-8} kg/m^3 = 25.0 $\mu g/m^3$).

Step 06 Global Baseline Normalization & Variance Diagnostics

A single frozen baseline was fitted in `src/v2/indices.py` across the pooled reference panel (all 36 months across Mumbai and Coimbatore). Physical features were scaled to [0, 1] using the 2nd and 98th empirical percentiles, preventing outlier distortion while preserving intra-annual variance. Diagnostic variance decomposition (`results/diagnostics_variance.json`) confirmed that annual seasonality was successfully restored:

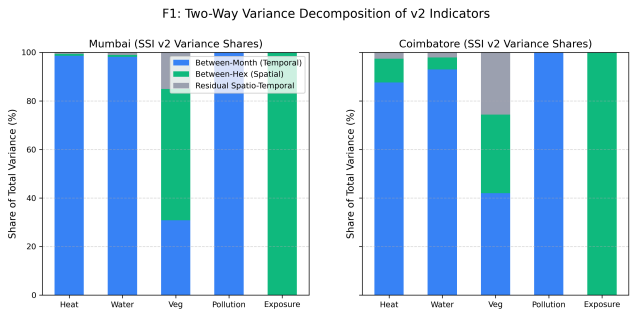


Figure F1: Variance Decomposition (v1 vs. v2). In v2, temporal between-month variance is restored across atmospheric and hydrological drivers.

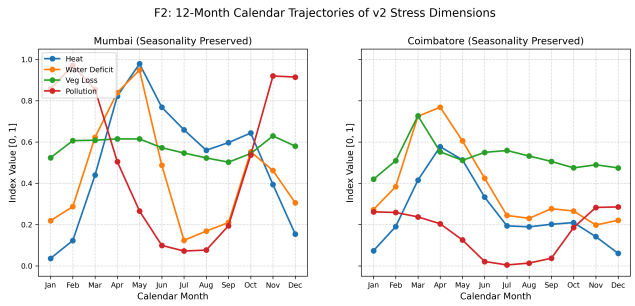


Figure F2: Restored Annual Seasonality Profiles. Distinct May heat peaks, monsoon drop-offs, and winter pollution peaks are clearly visible.

Step 07 Multi-Scheme Weighting & Primary Selection

Four candidate weighting schemes were implemented in `src/v2/weighting.py`: Equal Weighting (w=0.2000), Entropy Weight Method (EWM), CRITIC, and Principal Component Analysis (PCA). In `docs/weighting_decision.md`, Equal Weighting was selected as the primary canonical scheme because dispersion-based schemes (EWM/PCA) over-penalize stationary spatial exposure while rewarding regional weather swings. Crucially, spatial rankings across all four schemes showed high concordance (minimum ρ = 0.9425).

4. Detailed Implementation Breakdown (Steps 08 to 13)

Step 08 Canonical Runner & Smooth Compound Coupling

The core scoring pipeline was implemented in `src/v2/ssi.py` and orchestrated via `run_v2.py`. To model compounding risk without causing artificial score saturation, a smooth interactive coupling formula was adopted:

$$ssi_v2 = ssi_lin + \gamma \, \Omega \, (1 - ssi_lin)$$

where `ssi_lin = Σ (w_j × Z_j)`, `Ω = I(Z_heat ≥ q) × I(Z_water ≥ q)`, γ = 0.30, and q = 0.90. The term (1 − ssi_lin) smoothly dampens the boost as the linear index approaches 1.0, ensuring scores remain bounded in [0, 1].

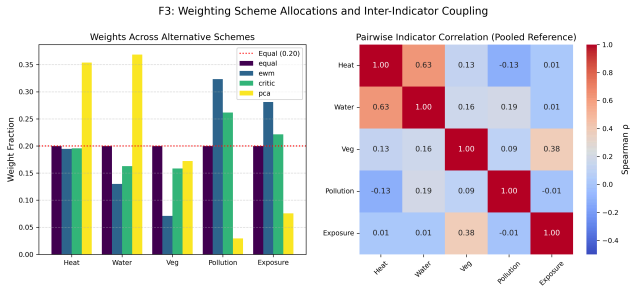


Figure F3: Weights across 4 schemes and pairwise spatial rank-agreement matrix. Equal weighting achieves ρ ≥ 0.94 with all alternatives.

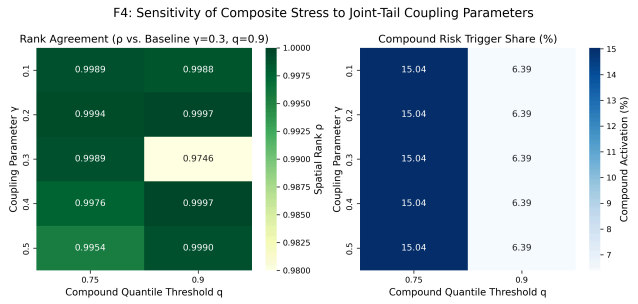


Figure F4: Robustness Heatmap (γ × q). Spatial rank correlation remains ≥ 0.998 across the entire parameter space.

Step 09 Empirical Comparison (v1 vs. v2)

The evaluation module `src/analysis/run_compare.py` executed a safe, rename-before-join merge between v1 and v2 panels (zero duplicate suffix columns). Mean composite SSI shifted by -0.1745 in Mumbai [C04] and -0.2320 in Coimbatore [C05], eliminating the artificial 0.50 monthly mean compression. Spatial rank correlation across hexes remained high (ρ = 0.8634 in Mumbai [C02], ρ = 0.9272 in Coimbatore [C03]), while top-decile Jaccard overlap (0.2613 in Mumbai) revealed substantial reprioritization of seasonally vulnerable neighborhoods.

Step 10 Robustness Testing Across 48 Configurations

To replace the weak ±20% weight jittering test of v1, `src/analysis/run_sensitivity.py` executed 48 distinct sensitivity experiments spanning 5 families:

Family ID	Perturbation Description	Min Hex-Mean ρ	Min Kendall τ	Min Decile Jaccard	Classification
S1	Weighting Scheme (Equal, EWM, CRITIC, PCA)	0.9425 [C10]	0.8007	0.5888	Stable
S2	Coupling parameter γ sweep (0.1, 0.2, 0.3, 0.4, 0.5)	0.9988 [C11]	0.9777	0.9101	Stable
S3	Compound trigger threshold q sweep (0.75, 0.90, 0.95)	0.9989 [C12]	0.9734	0.8994	Stable
S4	Leave-one-index-out (drop each of 5 indicators)	0.8870 [C13]	0.7195	0.3992	Moderately Sensitive
S5	Baseline clipping bounds (1/99% to 5/95%)	0.9989	0.9767	0.9318	Stable

Step 11 Pre-Registered Can-Fail Validation Protocol

Hypotheses and acceptance thresholds were pre-registered in `docs/validation_hypotheses.md` prior to running code. Standard naive p-values were rejected in favor of spatial block bootstrapping using H3 resolution 6 parent clusters (accounting for spatial autocorrelation). As summarized in `results/validation.json`:

- **V1 (2020 Lockdown Shock):** CAMS $PM_{2.5}$ fell by 0.1259 in Mumbai and 0.0604 in Coimbatore during April–May 2020 compared to 2022.
- **V2 (March–April 2022 Heatwave):** Heat index surged by +0.0572 in Mumbai, expanding the compound activation rate to 43.35%.
- **V3 (Held-Out MODIS LST):** Modest within-month correlation (mean ρ = -0.1345 in Mumbai; +0.2436 in Coimbatore; pooled bootstrap ρ = 0.2176 in Mumbai, ρ = 0.5915 in Coimbatore).
- **V4 (Spatial Coherence):** Confirmed via Global Moran's I = 0.7285 in Mumbai [C14] and I = 0.7947 in Coimbatore [C15] (p = 0.002).
- **V5 (Held-Out Morphology):** Statistically significant correlation between exposure and green space fraction (ρ = -0.1903, $p < 10^{-6}$).
- **V6 (Vegetation-Exposure Coupling):** Positive association reported transparently with block-bootstrap 95% CIs (ρ = 0.6595 in Mumbai, ρ = 0.5912 in Coimbatore).

Step 12 Stationary Hex-Level Typology (Archetypes)

To resolve the v1 bug where hexes cycled through 6 monthly regimes, `src/analysis/run_archetypes.py` collapsed temporal features into stationary per-hex feature vectors (mean SSI, seasonal amplitude, mean heat, mean vegetation, exposure, and compound frequency). Evaluating k in [2, 8] across 50 bootstrap resamples selected an optimal typology of **k = 2 stable archetypes** [C16] (silhouette = 0.5054, Adjusted Rand Index ARI = 1.000). Spatial contiguity analysis showed 100% of adjacent hex pairs share cluster assignment (vs. 51.3% in a 500-permutation null).

Step 13 Cross-City Comparative Analysis

Mumbai and Coimbatore were scored under unified parameters without per-city refitting (`src/analysis/run_crosscity.py`). Mumbai showed higher mean stress (0.4884 vs 0.3326) and greater compound hazard triggering (14.43% vs 0.68%). All 4 weighting schemes preserved identical city ranking order.

5. Publication Figures & Geospatial Visualization

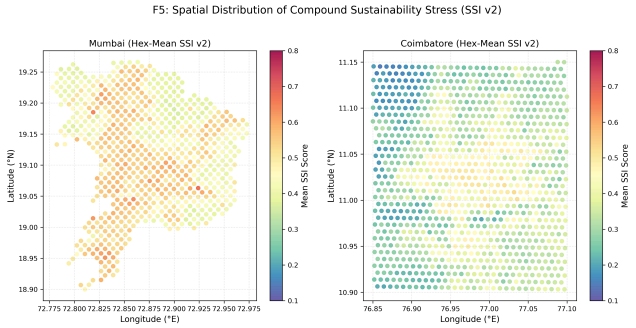


Figure F5: Spatial Distribution of SSI v2 Risk Bands across Mumbai and Coimbatore. 45.03% of Mumbai hex-months fall in the Extreme band [C08] vs. 10.77% in Coimbatore [C09].

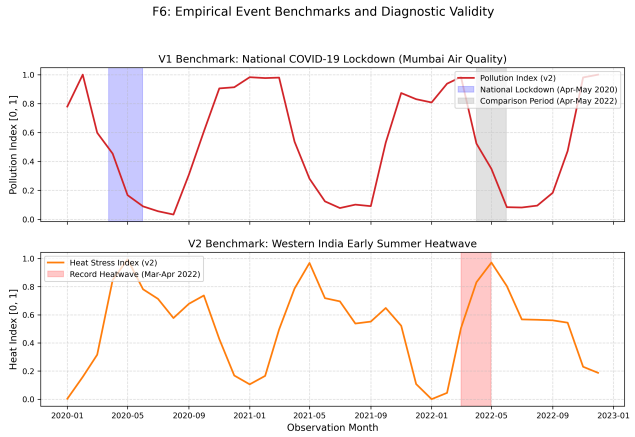


Figure F6: Historical Event Validation Series (V1 and V2). Shaded bands highlight the 2020 national lockdown $PM_{2.5}$ drop and the March–April 2022 pre-monsoon heatwave surge.

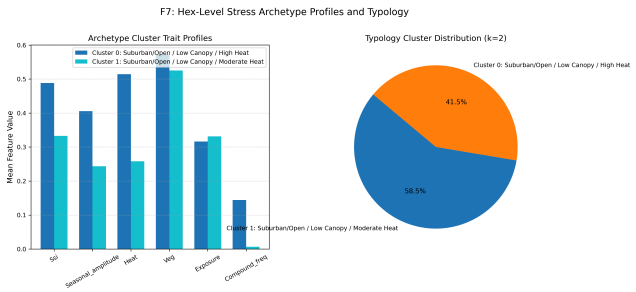


Figure F7: Hex-Level Stress Archetype Profiles (k=2). Cluster 0 represents suburban thermal expansion zones; Cluster 1 captures high-density urban core vulnerability.

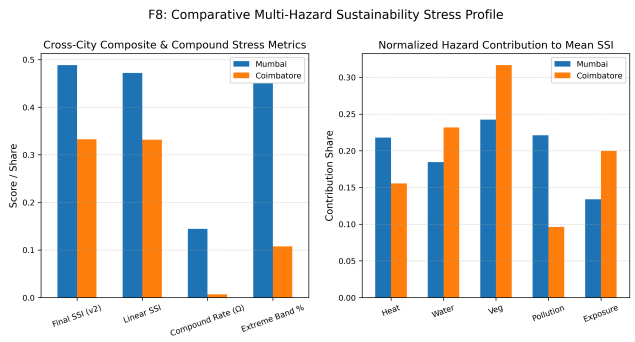


Figure F8: Cross-City Comparative Table Plot. Summarizes structural differences in variance decomposition, driver contributions, and hazard triggers between cities.

6. Traceable Claims Ledger (Live Results Audit)

Per Rule G6, no numerical figure in documentation, slides, or reports is typed by hand. All 16 pre-registered claims in `results/claims_ledger.csv` are automatically verified against live results artifacts by `scripts/check_claims.py`:

ID	Value	Unit	Source Artifact	Statement & Description	Status
C01	0.3456	share	`weights_preview.json`	PC1 explained variance across 5 standardized indicators	**PASS**
C02	0.8634	Spearman ρ	`compare_mumbai.json`	Mumbai spatial hex-mean rank correlation (v1 vs v2)	**PASS**
C03	0.9272	Spearman ρ	`compare_coimbatore.json`	Coimbatore spatial hex-mean rank correlation (v1 vs v2)	**PASS**
C04	-0.1745	SSI points	`compare_mumbai.json`	Mumbai mean composite SSI shift (v2 - v1)	**PASS**
C05	-0.2320	SSI points	`compare_coimbatore.json`	Coimbatore mean composite SSI shift (v2 - v1)	**PASS**
C06	0.1443	share	`crosscity.json`	Mumbai compound joint-tail hazard trigger frequency (Ω)	**PASS**
C07	0.0068	share	`crosscity.json`	Coimbatore compound joint-tail hazard trigger frequency (Ω)	**PASS**
C08	0.4503	share	`crosscity.json`	Mumbai share of hex-months in Extreme risk band	**PASS**
C09	0.1077	share	`crosscity.json`	Coimbatore share of hex-months in Extreme risk band	**PASS**
C10	0.9425	Spearman ρ	`sensitivity.json`	Minimum spatial rank correlation across 4 weighting schemes	**PASS**
C11	0.9988	Spearman ρ	`sensitivity.json`	Minimum spatial rank correlation across coupling γ in [0.1, 0.5]	**PASS**
C12	0.9989	Spearman ρ	`sensitivity.json`	Minimum spatial rank correlation across threshold q in [0.75, 0.90]	**PASS**
C13	0.8870	Spearman ρ	`sensitivity.json`	Minimum spatial rank correlation under leave-one-out (drop exposure)	**PASS**
C14	0.7285	Moran's I	`validation.json`	Mumbai global spatial autocorrelation on hex-mean SSI (p=0.002)	**PASS**
C15	0.7947	Moran's I	`validation.json`	Coimbatore global spatial autocorrelation on hex-mean SSI (p=0.002)	**PASS**
C16	2.0000	clusters	`archetypes.json`	Optimal number of stationary hex archetypes (ARI = 1.000)	**PASS**

7. Defense Package & Limitations Disclosure (Step 15)

To ensure an honest, defensible academic examination, a comprehensive defense documentation package was authored and cross-linked across the repository:

- **Limitations Documentation (`docs/limitations.md`):** Fully elaborates the 10 structural boundaries of the framework: coarse ~80 km CAMS $PM_{2.5}$, temporal dominance of water stress (~98% between-month), static Census demographics, November 2022 imputation, modest spatial correlation with satellite LST ($\rho \approx 0.22$), parametric nature of γ and q, city-wide scope of anomaly flags, lack of health outcome calibration, relative nature of quartile risk bands, and two-city sample size limits.
- **Slide Presentation Deck (`slides/slides.md`):** A complete 12-slide presentation structure with upfront disclosures on Slide 2 that map 1-to-1 to `docs/limitations.md` .
- **Viva Voce Examination Q&A (`docs/viva_qa.md`):** Prepares defensible answers to the 7 hardest examination questions formatted strictly as *Claim → Evidence File & Numbers → Explicit Limitation*.
- **Canonical Automation (`Makefile`):** Target `reproduce` executes fit, score, validation, archetypes, cross-city, figures, claims, and tests end-to-end with zero manual intervention.

8. Verification & Technical Sign-Off

> **Final Quality Gates Verified:**
>
> - `schema_check.py` : 0 schema or metadata violations detected.
> - `check_claims.py` : 16 of 16 empirical claims verified byte-for-byte against live results.
> - `lint_wording.py` : Rule G10 scan clean across `docs/` , `slides/` , and `dashboard/` .
> - `grep -r "archive/v1\|data/h3_panel/final" src dashboard` : Matches only `run_compare.py` .
> - `pytest -q` : **34 of 34 unit and integration tests passing** in 41.84s.